

CSW 439: Data Mining

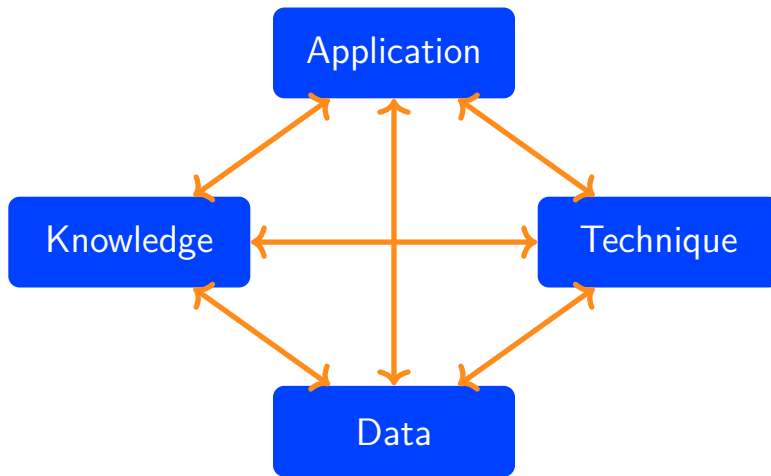
Lab 1: Data Mining Overview & Data Understanding

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Quarter 3, 2025-2026

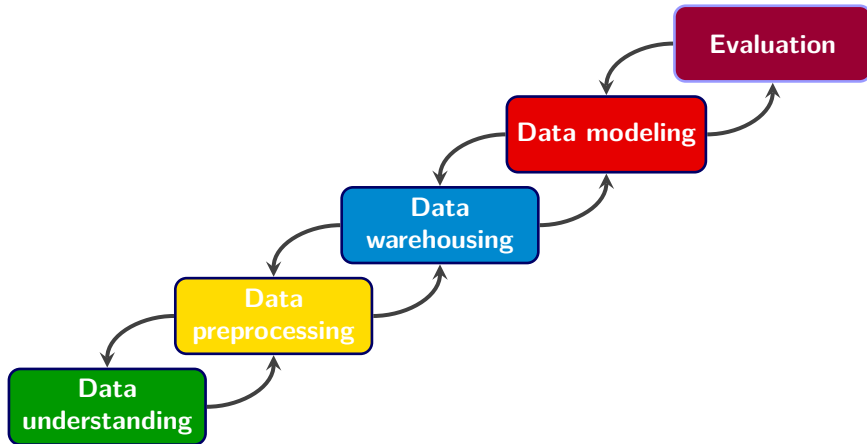
- **Big Picture of Data Mining:**
 - Four View
 - Data Mining Pipeline
- **Data Understanding:** The first stage of data mining pipeline.

Data Mining: Four Views



Reference: Diagram adapted from the course "Data Mining Foundations and Practice Specialization" by Qin (Christine) Lv (Coursera).

Data Mining Pipeline



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Data Mining Pipeline

1. Data Understanding

- Data Objects, attributes and statistics
- Data Visualization
- Data Similarity

2. Data Preprocessing

- Data Cleaning and Integration
- Correlation Analysis
- Data Normalization
- Discretization and Attribute selection
- Numerosity Reduction

3. Data Warehousing

- *(Skipped in this course)*

4. Data Modeling

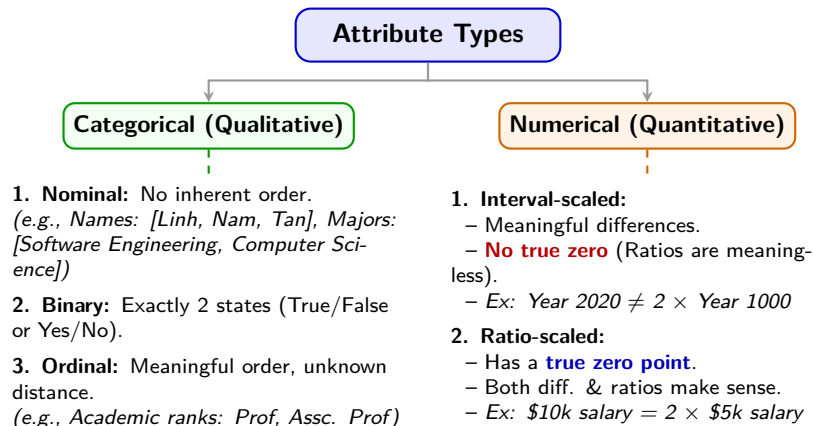
- Frequent Pattern Mining
- Clustering
- Classification
- Regression

5. Evaluation

- *Assessing model performance & business value*

Data Understanding: Data Objects & Attributes

- **Data Objects:** Represent real-world entities (e.g., employees, products). A dataset is a collection of these objects.
- **Attributes:** Features, variables, or dimensions describing the Data Objects.



Data Understanding: Data Statistics

1. Central Tendency

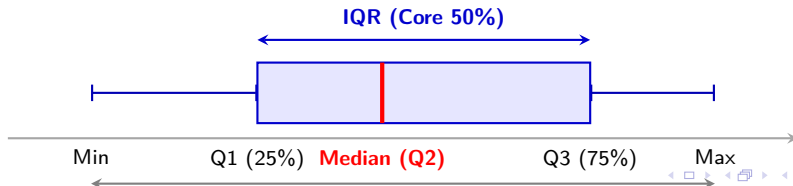
Where is the "typical" center?

- **Mean:** The average.
- **Median (Q2):** The exact 50% middle mark.
- **Mode:** Most frequent value.
- **Mid-range:** $\frac{Max + Min}{2}$.

2. Dispersion

How is the data spread out?

- **Range:** $Max - Min$.
- **Quartiles:** Q1 (25%), Q2 (50%), Q3 (75%).
- **IQR:** $Q3 - Q1$ (Covers the core 50%).
- **Variance & Standard deviation:**
Measure the degree of variability in the data.



The Dual Purpose of Visualization

Exploration

- **Goal:** Dig into the data to uncover *hidden patterns* and trends.
- **Action:** Quickly identify, investigate, and isolate anomalies.

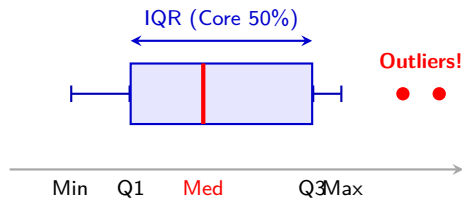
Explanation

- **Goal:** Translate complex findings into an *intuitive data story*.
- **Action:** Persuade non-technical decision-makers with visual evidence.

Data Visualization: Boxplot

Core Strengths:

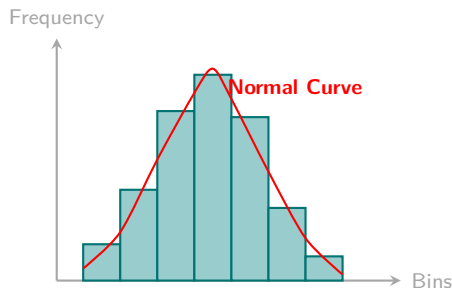
- Evaluates the distribution and establishes the **normal boundaries** of the dataset.
- **The Box (IQR)**: Encapsulates the core 50% of the data.
- **Catching Extremes**: Any point falling outside the safe threshold ($1.5 \times IQR$) is flagged and plotted individually.



Data Visualization: Histogram

Core Strengths:

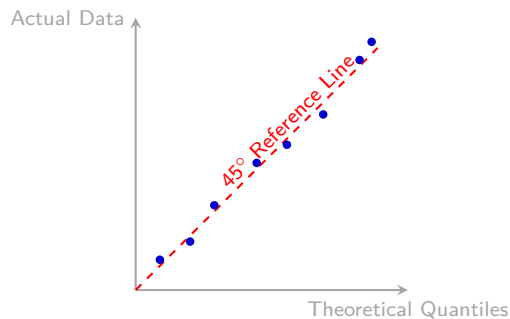
- Visualizes the overall shape and distribution of the entire range of data.
- **Mechanism:** Chops the value range into equal intervals (*bins/buckets*) and counts the frequency.
- **Identification:** Easily reveals if the data is bell-shaped (Normal), skewed, or multi-modal.



Data Visualization: Q-Q Plot

Core Strengths:

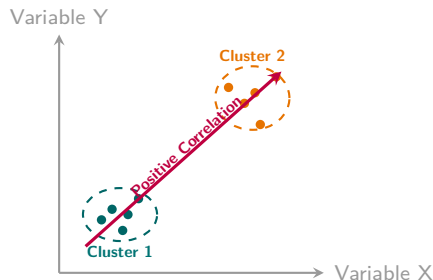
- Deeply verifies if the data truly follows a **theoretical normal distribution**.
- **How to read:** Don't look at the shape; look at the *alignment*.
- **Golden Rule:** If the points align on a **diagonal straight line** → The actual distribution perfectly matches the theory!



Data Visualization: Scatter Plot

Core Strengths:

- Analyzes **bivariate data** to understand the interaction between 2 variables (X and Y).
- **Correlation:** Instantly identifies positive (upward) or negative (downward) trends.
- **Clustering:** The human eye automatically detects natural point clouds, laying the groundwork for Segmentation algorithms.



Data Understanding: Data Similarity

Goal: How close or far apart are two data points?

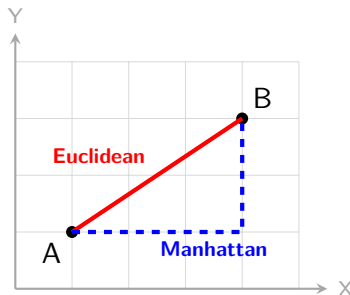
No single formula works for everything. We must match the metric to the data type.

1. Categorical & Ordinal

- **Nominal:** Match = 0, Mismatch = 1.
- **Binary:**
 - Symmetric → **Hamming** (counts bit flips).
 - Asymmetric → **Jaccard** (ignores "0-0" cases).
- **Ordinal:** Map to numbers, normalize $[0, 1]$, then subtract.

2. Numerical (Minkowski L_p Norm)

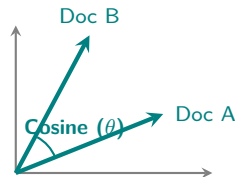
- $p = 1 \rightarrow$ **Manhattan:** Grid-like path.
- $p = 2 \rightarrow$ **Euclidean:** Straight physical line.



Data Understanding: Data Similarity

1. Text Data (Sparse Arrays)

- **Problem:** Physical distance fails when texts have different lengths.
- **Solution: Cosine Similarity.**
- Measures the *angle* (θ) between word vectors. Ignores document length, focuses purely on topic alignment.



2. Time Series Data

- **Problem:** Waves are similar in shape but delayed/out of phase.
- **Solution: Dynamic Time Warping (DTW).**
- Flexibly "warps" the time axis to match

